1. Select a mathematical description or a representation that describes a problem

The problem describes a real-world scenario: investing in a kite sail to save fuel costs. Students must **select a mathematical model**—specifically, a **payback period model**—by comparing the one-time investment with the annual fuel savings to calculate the time to recover the cost.

4. Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation

Students must **read and comprehend** the statements in the problem (diesel price, savings percentage, investment cost) and the chart (annual fuel consumption of the ship), and then **construct a mathematical model** that translates the real-world situation into mathematical calculations.

7. Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis

The real-world problem is complex (ship operations, fluctuating fuel prices, varying savings rates, etc.), but the question provides simplified assumptions (20% savings, fixed price). Students need to further **decompose the problem** into:

- Calculating total annual fuel costs
- Calculating annual savings
- Calculating the payback period (investment ÷ annual savings)

6. Identify and describe the mathematical aspects of a real-world problem situation including identifying the significant variables

Students must identify key variables:

- Diesel price (0.42 zeds/L)
- Annual fuel consumption (read from the chart)
- Savings rate (20%)
- Investment cost (2,500,000 zeds)

9. Translate a problem into a standard mathematical representation or algorithm

The problem needs to be translated into **standard mathematical expressions**:

$$\text{Annual Savings} = \text{Annual Fuel Consumption} \times 0.42 \times 20\%$$

$$\text{Payback Period} = \frac{2,500,000}{\text{Annual Savings}}$$

2. Identify the key variables in a model

Similar to point 6, but more focused on clarifying the role of variables within the established model.

1. Select a mathematical description or a representation that describes a problem

题目描述了一个实际问题：投资风筝帆以节省燃油费用。学生需要**选择数学模型**——即“投资回收期”模型，通过比较一次性投资与年度节省费用来计算回收时间。

4. Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation

学生必须**阅读理解**题目中的陈述（柴油价格、节约百分比、投资成本）和图表（船年耗油量），并据此**建立一个数学模型**，将现实问题转化为数学计算。

7. Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis

现实问题复杂（船只运行、燃油价格波动、节约比例等），但题目已提供简化假设（节约20%，价格固定）。学生需在此基础上进一步**分解问题**为：

- 计算年燃油总成本
- 计算年节约金额
- 计算回收期（投资 ÷ 年节约）

6. Identify and describe the mathematical aspects of a real-world problem situation including identifying the significant variables

学生需识别关键变量：

- 柴油价格（0.42 zeds/L）
- 年耗油量（从图中读取）
- 节约比例（20%）
- 投资成本（2,500,000 zeds）

9. Translate a problem into a standard mathematical representation or algorithm

问题需转化为**标准数学表达式**：

$$\text{年节约} = \text{年耗油量} \times 0.42 \times 20\%$$

$$\text{回收期} = \frac{2,500,000}{\text{年节约}}$$

2. Identify the key variables in a model

与第6条类似，但更侧重于在已建立的模型中明确变量角色。